

D I V E R G E N T R E S O U R C E L O G I C — T E C H N I C A L R E P O R T

Is GWP Solving Anything?

An Audit of the Zero-Baseline Rule in Global Warming Potential Accounting

A technical report identifying the specific standards committees, decisions, and dates responsible for treating a standing forest's pre-extraction climate function as zero in global warming potential (GWP) accounting — the same convention that correctly assigns mined ore a near-zero pre-extraction value. The report asks a methodological question first — does GWP accounting, as currently applied, verify climate outcomes or merely enable material comparison — then establishes who made the relevant boundary decision and when, states why the resulting omission cannot be expressed as a single GWP-compatible number, quantifies the portion that can be measured using only published, peer-reviewed methodology, and compares the resulting global-scale figure against the annual emissions of the global steel and cement industries.

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EXECUTIVE SUMMARY

What this report establishes

Global warming potential (GWP) accounting, as applied to structural construction materials, assigns each material a pre-extraction baseline. For mined materials — steel, aluminium, concrete's mineral inputs — that baseline is correctly zero: ore in the ground does not perform a comparable ongoing atmospheric carbon-removal function. For forest biomass, the same accounting convention also assigns a pre-extraction baseline of zero. That zero is false. A standing forest performs a continuous, measurable, peer-reviewed climate function before any harvest occurs: net carbon removal, evapotranspirative cooling, and (region-dependent) radiative forcing effects. Setting this baseline to zero is not a simplification of a small effect. In the systems analysed by the literature this report relies on (Sections 3–5), it is the omission of the largest quantified term in the calculation.

This report does four things. First, it identifies who set the accounting boundary that produces the zero, what they decided, when, and what alternative was available and not taken — because a convention chosen by a specific body at a specific date, rather than a law of physics, is a convention that can be challenged on the merits. Second, it states the resulting accounting error precisely, in the vocabulary of GWP itself, so the claim can be checked against the standard rather than argued rhetorically. Third, it quantifies the omitted term using only published, peer-reviewed methodology — principally Maxwell et al. (2019, *Science Advances*), which already built and validated a forgone-removals accounting model for tropical forest loss, finding that full accounting increases the net carbon impact of intact forest loss by 626% relative to direct-clearance emissions alone. Fourth, it compares the resulting global-scale figure against the published annual CO₂ emissions of the global steel and cement industries, to test whether full-boundary forestry carbon impact is comparable to, or larger than, entire industrial sectors currently subject to far heavier disclosure and regulatory scrutiny.

Headline finding: Using the Maxwell et al. (2019) forgone-removals multiplier (626%) applied to the global net forest sink reported by the World Resources Institute (–5.1 GtCO₂e/year, 2001–2025 average) and IPCC AR6 methane GWP-100 values for the slash/decomposition pathway already documented in the DRL Codex, the full-boundary climate impact of global annual forest loss and degradation is of a magnitude comparable to, and by some published regional estimates exceeding, the combined annual direct emissions of the global steel industry (2.6–2.8 GtCO₂/year, IEA) and the global cement industry (≈1.6–2.5 GtCO₂/year gross, IEA/GCCA). Forestry's climate impact is not a rounding error against these sectors. On a full-boundary accounting basis it is the same order of magnitude as both combined — and unlike steel and cement, it is not currently subject to comparable mandatory disclosure.

The report does not propose a single new GWP-compatible number for "cutting a forest," because no such single number can exist under the current metric's own definition — this is argued formally in Section 2. Instead, it identifies the specific body,

decision, and date responsible for the omission, sources every figure used to peer-reviewed literature or primary government data, and shows the work required to reproduce the headline comparison from public sources.

SECTION 0

Who, What, When, Where, and Why: The Origin of the Zero

Before the mathematics, the convention responsible for it deserves to be named precisely. A recurring response to this report's central claim will be that the zero-baseline treatment of forest biomass is simply how carbon accounting works — an inevitability of the science, not a choice. The record does not support that response. The zero was a specific decision, made by a specific body, at a specific date, with a specific available alternative on the table at the time. This section establishes that record before any of the quantitative work that follows.

0.1 What was decided

The accounting rule under examination is formally known as the biogenic carbon characterisation factor in ISO 21930, the international standard governing Environmental Product Declarations (EPDs) for construction products. The rule assigns a characterisation factor of -1 to biogenic carbon entering a product system (i.e., carbon absorbed by the tree as it grew is counted as a negative emission, a credit) and $+1$ when it leaves the system (combustion or decomposition). This is the " -1 in / $+1$ out" convention. Critically, the standard further provides that wood may claim the -1 credit only "when the wood originates from sustainably managed forests" — and assigns a characterisation factor of 0 (not 1) to land-use and land-use-change emissions when the forest of origin is certified as sustainably managed. An unsustainably managed forest retains a characterisation factor of 1 ; a certified forest is reset to 0 . It is this second figure — the LULUC characterisation factor reset to zero upon certification — that is the specific mechanism this report has been calling "the zero" throughout.

0.2 Who decided it

ISO 21930:2017 was developed by ISO Technical Committee 59, Subcommittee 17, Working Group 3 (ISO/TC59/SC17/WG3). Its European counterpart, EN 15804, was developed by the corresponding CEN technical committee (CEN/TC350/WG3). These are technical standards committees, not elected or government mitigation bodies; their formal mandate is harmonised measurement methodology, not climate policy. Publicly disclosed committee participation includes experts seconded from national standards bodies, government laboratories, and industry trade associations — a structure common to ISO and CEN technical committees generally, across all material sectors, and not specific to forestry. One identifiable case, disclosed in industry literature reviewed for this report: an expert credited as the named author of an explanatory technical paper on timber carbon accounting served simultaneously as

the UK expert on both CEN/TC350/WG3 (which developed EN 15804:2011) and ISO/TC59/SC17/WG3 (which developed ISO 21930:2017) — the two committees responsible for the parallel European and international standards under examination in this report — while also affiliated with Timber Development UK, a trade body formed from the merger of the Timber Trade Federation and the Timber Research and Development Association.

This is disclosed here as a documented configuration, not as evidence of improper conduct by any individual or institution. Dual expert/industry roles are standard, permitted practice in voluntary consensus standards development, and the same structure exists for steel, cement, and aluminium standards committees. No claim is made in this report that any participant acted to produce a particular outcome, that the outcome was foreseen by any participant, or that the standard-setting process departed from normal ISO/CEN procedure. The relevant and narrower observation is structural, not personal: technical expertise relevant to a product category's own measurement standard is, by the ordinary operation of voluntary consensus standards-setting, disproportionately supplied by participants with professional grounding in that product category. This is true of every ISO and CEN technical committee reviewed for this report, in every material sector. It is noted here only because it is the documented composition of the specific committees that produced the specific boundary decision this report examines, and a reader assessing the boundary decision is entitled to know who was disclosed as present when it was made.

0.3 When it was decided, and what alternative existed at the time

ISO 21930 was first published in 2007 and substantially revised in 2017; EN 15804 was first published in 2012. By the time of the 2017 ISO revision, the foundational peer-reviewed literature on forest carbon-sink dynamics that this report relies on in Sections 3–5 was already in print: Pan et al. (2011, Science) had already established the global forest carbon sink synthesis; Stephenson et al. (2014, Nature) had already published the finding that carbon accumulation rate increases continuously with tree size, meaning the standard's drafters had access to peer-reviewed evidence that older, larger — i.e., more commercially valuable — trees disproportionately perform the removal function the standard was about to set to zero. The "–1 in / +1 out with certification reset to 0" convention was not, therefore, adopted in the absence of available science describing the removal function being zeroed out. The available alternative — incorporating a forgone-removals or counterfactual-baseline term, structurally analogous to the method Maxwell et al. would later formalise and publish in 2019 — was not selected.

0.4 Where the consequence lands

The boundary decision is European and international in its standard-setting origin (CEN, Brussels; ISO, Geneva) but its consequence is realised wherever the standard is adopted into national procurement and disclosure law: in the United States, through USDA grant programme criteria and the 2023 Buy Clean Initiative's EPD-based procurement preference (documented in the DRL companion Reagan Forestry Legacy paper, Section 6.2); in the European Union, through EN 15804-conformant EPDs required under national green-building codes; and in every jurisdiction where a building's claimed embodied-carbon figure is calculated by a WoodWorks- or equivalent industry-distributed calculator defaulting to the standard's own –1/+1

convention. The standard-setting decision and its procurement consequence are geographically and institutionally distinct, which is itself part of why the boundary choice receives less public scrutiny than a domestic legislative act would: by the time it reaches a procurement officer or a building owner, it presents as settled international technical consensus rather than as a contestable choice made by a working group with disclosed industry participation.

0.5 Why: the test this report applies

This report does not assert that the committees responsible for ISO 21930 or EN 15804 intended the outcome documented in Sections 3 through 5. Consistent with the evidentiary standard applied throughout the DRL corpus, intent is not characterised; configuration is described. The relevant test is not "did anyone intend this," but a narrower and answerable one: was the available science, at the time of the decision, sufficient to know that the zero-baseline convention would not be accurate for forest biomass in the way it is accurate for inert materials? Section 0.3 establishes that the answer is yes — Pan et al. (2011) and the broader forest-sink literature predate the 2017 ISO revision. A standards committee is not obligated to anticipate science that does not yet exist. This committee was not in that position. The science existed. The convention was adopted, and has not since been revised to incorporate it, notwithstanding that Maxwell et al. (2019) subsequently formalised exactly the correction the existing literature already made available.

0.6 A second, earlier instance of the same pattern: the IPCC's own warning, dropped downstream

Section 0.1 through 0.5 examine a single boundary decision, made by named ISO and CEN committees in a specific period (2011–2017). It is worth distinguishing that decision from an earlier, separate event in the same accounting lineage, because the two are sometimes conflated and the distinction matters to an accurate account of how the omission this report documents actually arose.

The biogenic-carbon accounting convention used throughout construction-material EPDs — carbon absorbed by a tree treated as a credit, carbon released at combustion or decomposition treated as a debit — originates in the IPCC's 1996 and 2006 Guidelines for National Greenhouse Gas Inventories, specifically the stationary combustion accounting structure (IPCC 2006, Volume 2, Chapter 2; Eggleston, Buendia, Miwa, Ngara & Tanabe, eds., IGES). The rule exists there for a narrow and defensible bookkeeping reason, unrelated to forestry advocacy: it prevents a tonne of biogenic carbon from being counted twice, once in the energy sector at the point of combustion and again in the land-use (AFOLU/LULUCF) sector at the point of harvest. Biogenic combustion emissions are reported as an "uncounted informational item" in national energy-sector totals specifically so the same carbon is counted once, in the land-use sector, not because the carbon is treated as climatically inconsequential.

Critically, the IPCC's own guidance has repeatedly and explicitly warned against the reading that later became standard industry and regulatory practice. The IPCC Task Force on National Greenhouse Gas Inventories has stated directly that this sector-boundary convention "should not be interpreted as a conclusion about the sustainability, or carbon neutrality of bioenergy," and that "IPCC Guidelines do not automatically consider or assume biomass used for energy as 'carbon neutral,' even in

cases where the biomass is thought to be produced sustainably" (IPCC Task Force on National Greenhouse Gas Inventories, 2024, as cited in Booth, M.S., "Burning Up the Carbon Sink: How the EU's Forest Biomass Policy Undermines Climate Mitigation," GCB Bioenergy, 2025).

The accounting convention's own source body has, on the public record, repeatedly disclaimed the conclusion that the convention is now used to support. A bookkeeping rule built to prevent double-counting between two sectors of a national inventory was, downstream, converted into a public-facing claim of "carbon neutral" — a claim the rule's own authors say it was never intended to establish.

This is a different mechanism from the terminology sequence documented in the DRL companion Reagan Forestry Legacy paper, Section 5 (the "CLT" → "mass timber" → "innovative wood products" rebrand, 2014–2026, which carried an existing accounting treatment into new federal procurement and tax-code categories without changing the underlying convention). The IPCC sector-boundary convention was not renamed; it was reinterpreted, with the IPCC's own disclaimer left behind in the technical guidance documents rather than carried forward into the public-facing standards (ISO 21930, EN 15804) and industry tools (WoodWorks Carbon Calculator) built on top of it. Taken together, the two events show the same general pattern — an accounting or terminology choice, made for a narrow and defensible technical reason, subsequently carried into downstream policy and procurement instruments in a form that exceeds what the originating body stated or intended — occurring twice, at two different layers of the accounting stack, roughly two decades apart.

Section 0.6 finding: The IPCC's own technical guidance has, on the public record, repeatedly stated that its biogenic-carbon sector-boundary convention does not establish or assume carbon neutrality for biomass, "even in cases where the biomass is thought to be produced sustainably." Downstream construction-material standards and industry tools have, since at least 2017 (Section 0.3), built a "carbon neutral" public claim directly on top of that convention without carrying forward the originating body's own disclaimer. This report does not characterise that omission as intentional on the part of any committee or institution; it records that the disclaimer exists, in the public record, predating the standards examined in this report, and that it does not appear in the standards' public-facing characterisation-factor rules examined in Sections 0.1–0.4.

Sources: IPCC (2006). 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Chapter 2 (Stationary Combustion). Eggleston, H.S., Buendia, L., Miwa, K., Ngara, T. & Tanabe, K. (eds). IGES, Japan. IPCC (1997). Revised 1996 IPCC Guidelines for National Greenhouse Gas Inventories. IPCC/OECD/IEA, Paris. IPCC Task Force on National Greenhouse Gas Inventories (2024), as cited in Booth, M.S. (2025), "Burning Up the Carbon Sink: How the EU's Forest Biomass Policy Undermines Climate Mitigation," GCB Bioenergy. DOI: 10.1111/gcbb.70035.

Section 0 finding: The zero-baseline treatment of forest biomass is not an inevitable feature of carbon science. It is, on the evidence assembled in this section, the product of two distinct events: an IPCC sector-boundary bookkeeping convention (1996/2006) whose own source body has repeatedly disclaimed the "carbon neutral" reading later built on top of it, and a specific characterisation-factor decision by named ISO and CEN technical committees (finalised 2017 and 2011/2012 respectively, with disclosed industry trade-

association participation) that adopted the zero-baseline rule at a date by which the peer-reviewed forest-sink literature already contained the evidence needed to know the convention was inaccurate for the material category it governs. The remainder of this report quantifies the scale of the gap these decisions produced.

Sources: ISO 21930:2017, Sustainability in buildings and civil engineering works – Core rules for environmental product declarations of construction products and services. EN 15804:2012+A2:2019, CEN. Committee composition: ITC-SA technical paper, "Assessing carbon impacts of timber" (Anderson, UK expert, CEN/TC350/WG3 and ISO/TC59/SC17/WG3; Timber Development UK). WoodWorks, "When to Include Biogenic Carbon in an LCA," citing ISO 21930:2017 -1/+1 and sustainably-managed-forest certification provisions. Pan, Y. et al. (2011). Science 333(6045), 988–993. Stephenson, N.L. et al. (2014). Nature 507, 90–93.

SECTION 1

The Question, Restated Precisely

The colloquial question — "what is the GWP of cutting a forest?" — cannot be answered with a single number, for a structural reason explained in Section 2. But the question can be restated in a form that GWP accounting is built to answer, and that restated question is the one this report answers:

What pre-extraction baseline does current carbon accounting assign to forest biomass, is that baseline accurate, and if not, what is the scale of the resulting omission when the same accounting standard is applied consistently across material categories?

This reframing matters because it converts a philosophical objection into a testable accounting claim. The claim under test is narrow and falsifiable: that the baseline-zero convention, applied identically to inert mined materials and to living forest ecosystems, produces an accurate result for the former and an inaccurate result for the latter, and that the size of that inaccuracy is measurable using data already published in the peer-reviewed literature.

SECTION 2

Why GWP Cannot Produce a Single Number for Forest Loss

Global warming potential is a unit defined for comparing the radiative forcing of a fixed mass of a specific gas, integrated over a specific time horizon, against the radiative forcing of an equivalent mass of CO₂ over the same horizon (IPCC AR6 WG1, Chapter 7). It requires three defined inputs: a gas, a mass, and a time horizon. It does not have a native unit for the cessation of an ongoing biophysical process.

Felling timber and mining ore are not symmetrical events under this definition. Ore extraction begins from a baseline at or near zero for the function this report is concerned with: rock in the ground does not perform an ongoing, measurable carbon flux, evapotranspiration, or radiative forcing function comparable to a living forest canopy prior to disturbance. (Ore bodies are not climatically inert in every sense — extraction affects local hydrology, geology, and soil chemistry — but they do not perform the specific atmospheric removal function under examination here.) Every GWP-relevant emission associated with steel or aluminium production is attributable to the extraction and processing activity itself — there is no antecedent atmospheric-removal service being withdrawn. Forest clearance is different in kind. Immediately prior to harvest, the stand was performing three concurrent, independently measurable functions:

Carbon flux — net ecosystem exchange (NEE) measured by eddy covariance towers (FLUXNET/AmeriFlux networks) shows mature forest stands operating as continuous net carbon sinks, with documented sign-flip to net carbon source within the same growing season following clearcut (Williams et al. 2014, Global Change Biology, DOI 10.1111/gcb.12388).

Evapotranspirative (latent heat) flux — forest canopy converts absorbed solar energy primarily into water vapour rather than sensible (surface) heat, a function lost upon clearance (Luyssaert et al. 2014, Nature Climate Change, DOI 10.1038/nclimate2196).

Radiative forcing via albedo — a smaller, regionally and seasonally variable term, partially offsetting (1) and (2) in snow-affected temperate/boreal zones, additive to (1) and (2) in tropical zones (Luyssaert et al. 2014; Bright et al. 2015).

GWP, by its IPCC definition, has a native unit for none of these three functions in their pre-extraction, ongoing form. (1) is a flux rate (tCO₂/ha/year), not a mass of emitted gas. (2) and (3) are measured in watts per square metre (radiative forcing / energy flux), an entirely different physical unit with no fixed conversion to a GWP-compatible CO₂e figure without an explicit, debated set of modelling assumptions (Bright & Strømman 2014; Cherubini et al. 2012). This is not a gap current methodology happens not to have filled. It is a structural mismatch between the metric and the object being measured.

Conclusion of Section 2: There is no single GWP-denominated figure for "cutting a forest" because two of the three lost functions are not expressed in GWP's native units and cannot be converted to those units without methodology choices that are themselves actively contested in the peer-reviewed literature. This is true. It is also, on its own, the strongest evidence for the report's central claim: a metric that cannot account for what a forest does is being used, unmodified, to certify forest-derived materials as low-carbon — not because the accounting is complete, but because the parts it cannot express are silently treated as zero.

Section 3 isolates the one component of the three — carbon flux — that does convert cleanly to GWP-compatible units, and quantifies it using only published, peer-reviewed methodology. The evapotranspiration and albedo components are documented qualitatively in Section 2 above and flagged in Section 6 as requiring separate, non-GWP metrics; they are not forced into the headline number, because doing so would replicate the same methodological overreach this report is built to expose.

SECTION 3

Quantifying the Carbon-Flux Component: The Forgone Removals Method

3.1 The published precedent

The accounting method this report applies is not novel. It was built, peer-reviewed, and published by Maxwell et al. (2019), "Degradation and forgone removals increase the carbon impact of intact forest loss by 626%," *Science Advances* 5(10), eaax2546, DOI 10.1126/sciadv.aax2546. The authors' stated objective was explicit: standard emissions accounting for tropical deforestation counts only the direct carbon released at clearance, and "is not considered in conventional emission accounting frameworks" — their words — the forgone carbon sequestration that the cleared area would otherwise have continued to perform.

3.2 The published formula

Maxwell et al. define forgone removals (F_n) for a given region j and year n as:

$$F_n = E_j \times R_{jn}$$

where E is the total area (hectares) in lost parcels in region j subject to clearance, selective logging, or edge effects, and R is the intact forest carbon sequestration rate (tonnes carbon per hectare per year) for that region in year n , drawn from independently published sequestration-rate data (Pan et al. 2011, the primary global forest carbon sink synthesis cited by Maxwell et al.).

3.3 The published result

Applying this formula to pantropical intact forest loss between 2000 and 2013, with forgone removals and degradation effects (selective logging, edge effects) projected through 2050, Maxwell et al. report:

Accounting component	Carbon impact (Tg C, Pantropics)	95% CI
Direct clearance emissions (2000–2013) — conventional accounting baseline	338	372–208
Degradation emissions (selective logging, edge effects, 2013–2050)	806	878–465
Forgone carbon removal (2000–2050)	972	1,604–331
Full-accounting net total	2,116	2,854–1,004

Source: Maxwell et al. (2019), Table 1 and Fig. 2. Tg C = teragrams (million tonnes) of carbon. To convert to CO₂e, multiply by 3.667 (the molecular weight ratio of CO₂ to C).

The conventional-accounting baseline (338 Tg C) captures only direct clearance emissions. The full-accounting total (2,116 Tg C) is 626% larger. The single largest component of the omitted total is not the degradation term — it is forgone removals (972 Tg C), the carbon that the cleared and degraded area would have continued to sequester had it remained intact. This is the empirical, peer-reviewed confirmation of the mechanism this report is built around: the carbon-flux component of forest loss, honestly accounted, is dominated by the term that current convention sets to zero.

3.4 Why the omitted term compounds rather than stays flat

The forgone-removals term is not a flat annual figure — it is weighted toward the highest-sequestration-rate hectares, which the peer-reviewed record shows are disproportionately the oldest, largest trees: Stephenson et al. (2014, *Nature* 507, pp. 90–93, DOI 10.1038/nature12914) analysed 673,046 trees across 403 species in six countries and found that "for most species mass growth rate increases continuously with tree size," such that a single large tree can sequester as much carbon in one year as an entire medium-sized tree gains over the same period. At one study site, trees in the largest size class comprised 6% of stems but accounted for 33% of annual stand-level growth. This is also the segment of the forest with the highest commercial timber value, meaning the harvest-priority order and the sink-value-priority order point at the same trees.

Sources: Maxwell, S.L. et al. (2019). *Science Advances* 5(10), eaax2546. DOI: 10.1126/sciadv.aax2546. Stephenson, N.L. et al. (2014). *Nature* 507, 90–93. DOI: 10.1038/nature12914. Pan, Y. et al. (2011). "A Large and Persistent Carbon Sink in the World's Forests." *Science* 333(6045), 988–993.

SECTION 4

Scaling to a Global Annual Figure

4.1 The global forest sink baseline

The World Resources Institute Global Forest Review, drawing on Harris et al. (2021, *Nature Climate Change*, DOI 10.1038/s41558-020-00976-6) and updated satellite-based emissions/removals modelling, reports that between 2001 and 2025, average annual gross emissions from deforestation and forest disturbance were 9.3 GtCO₂e/year, against average annual gross removals by forests of –14.4 GtCO₂e/year. The net sink figure used throughout this report, –5.1 GtCO₂e/year, is the published difference between these two figures (–14.4 + 9.3 = –5.1), described by WRI as equivalent to "the world's forests removing the annual GHG emissions of India and Japan" combined. Readers seeking to verify this figure directly should consult the WRI Global Forest Review "Greenhouse Gas Fluxes from Forests" indicator at gfr.wri.org, which publishes the underlying gross emissions and gross removals series from which the net figure is derived.

This net sink figure is the conventional-accounting baseline: it already nets gross removals against gross emissions, in the same manner Maxwell et al. describe as standard practice. It does not include a forgone-removals correction.

4.2 Defining “full-boundary” for this report

Full-boundary accounting, as the term is used in this report, means: direct emissions from forest loss and degradation, plus the forgone carbon removals that would have occurred had the affected area remained intact. This is distinct from, and additional to, the three building-level liabilities (soil organic carbon efflux, end-of-life methane, foregone sequestration) documented elsewhere in the DRL Codex; this report’s figure is a global-scale estimate of the forgone-removals term specifically, modelled on the Maxwell et al. (2019) method.

4.3 Applying the forgone-removals multiplier — corrected derivation

Maxwell et al.’s 626% full-accounting multiplier (6.26×) was derived specifically for pantropical intact forest loss — a higher-sequestration-rate, higher-biomass forest type than the global average, which includes substantial lower-density boreal and temperate secondary forest and plantation harvest. The 6.26× figure was published by its authors as a finding for that specific forest type and region; it was not presented by Maxwell et al. as a universal global coefficient, and this report does not treat it as one. Applying it without adjustment to the global figure would overstate the case. This report instead applies a conservative, self-imposed discount — a 2.0–3.0× multiplier, roughly one-third to one-half of the published pantropical figure — stated here explicitly as a modelled estimate selected by this report’s author to avoid the over-claiming this report is structured to call out in others’ work, not as a separately validated global coefficient drawn from the literature.

The multiplier is applied to the forgone-removals component, not to direct emissions. The forgone-removals component is derived from the net forest sink (–5.1 GtCO₂e/year, itself the difference between gross removals of –14.4 Gt and gross emissions of 9.3 Gt, per WRI/Harris et al. 2021). The calculation proceeds in two steps:

Step 1 – Forgone removals: net sink (–5.1 GtCO₂e/yr) × [2.0–3.0] = –10.2 to –15.3 GtCO₂e/yr forgone removals

Step 2 – Full-boundary total: direct emissions (9.3 Gt) + forgone removals (10.2–15.3 Gt) = 19.5–24.6 GtCO₂e/yr

This corrected derivation — multiplying the net-sink-derived forgone-removals term, then adding direct emissions, rather than multiplying direct emissions alone — is the basis for the comparison in Section 5. (An earlier internal draft of this report multiplied direct emissions directly; that calculation conflated a flow term, the annual net sink, with the cumulative, multi-decade forgone-removals projection in Maxwell et al., and has been corrected here. The resulting range, 19.5–24.6 GtCO₂e/year, is closely comparable in magnitude to the originally stated 18.6–27.9 Gt range and does not change this report’s headline finding or its comparison to industrial sectors in Section 5, but the corrected derivation is the one this report now relies on throughout.)

Section 4 finding: Even using a self-imposed discount of roughly one-third to one-half of Maxwell et al.’s own published, peer-reviewed pantropical figure, and even after correcting the arithmetic to apply the multiplier to the forgone-removals term rather than to direct emissions, full-boundary forest-loss carbon impact (19.5–24.6 GtCO₂e/year) is more than double the figure currently used

in national GHG inventories and EPD-based material carbon accounting (9.3 GtCO₂e/year). The gap itself — at minimum, roughly 10 GtCO₂e/year of currently unrecognised annual climate impact — is larger than the entire global steel industry’s annual emissions on its own (Section 5). This figure is a modelled estimate built on a published, peer-reviewed regional method and a stated, conservative discount; it is not a directly measured global quantity, and Section 6 states this limitation explicitly.

SECTION 5

The Industrial Comparison

To test whether forestry’s full-boundary climate impact is comparable in scale to industries that face substantially heavier disclosure, regulatory, and public scrutiny, this section places the Section 4 figures alongside the most recent published annual emissions totals for the global steel and cement industries — the two structural-material sectors most directly in competition with timber, and the subject of the DRL companion materials-timeline analysis.

Sector	Annual CO ₂ / CO ₂ e (Gt/year)	Source	Accounting basis
Global steel industry	2.6–2.8	IEA, Iron and Steel Technology Roadmap; Net-Zero Industry Tracker 2024	Direct (Scope 1) industry emissions
Global cement industry	1.6 (process) / ≈2.5 (gross, incl. fuel + electricity)	IEA Cement; Global Efficiency Intelligence; GCCA	Direct process + fuel + indirect electricity
Combined steel + cement	≈4.2–5.3	As above	Sum of disclosed direct industry emissions
Forestry — conventional accounting (current practice)	9.3 gross / –5.1 net	WRI Global Forest Review (Harris et al. 2021 methodology)	Gross emissions netted against gross removals
Forestry — full-boundary, conservative floor (this report)	19.5–24.6 gross	This report, §4.3: net sink × [2.0–3.0] + direct emissions	Direct emissions + forgone removals
Forestry — full-boundary, Maxwell et al. published pantropical rate, unadjusted	≈41.2 (9.3 + 5.1×6.26)	Direct application of Maxwell et al. (2019) 626% multiplier to net sink, no discount applied	Direct emissions + forgone removals, unadjusted

Steel: IEA (2023, 2024). Cement process emissions: IEA Cement; cumulative process emissions reached 1.58 GtCO₂/year in 2023 (Copernicus ESSD 2025, citing Andrew 2019

cement emissions series). Forestry baseline: WRI Global Forest Review, Harris et al. (2021) methodology, 2001–2025 average.

Read conservatively — using only the self-imposed discount range from Section 4.3, not Maxwell et al.’s own higher published pantropical rate — full-boundary forestry climate impact (19.5–24.6 GtCO₂e/year) is roughly three-and-a-half to nearly six times larger than the combined disclosed direct emissions of the global steel and cement industries ($\approx$ 4.2–5.3 GtCO₂/year). Applying Maxwell et al.’s own published pantropical multiplier to the net sink without discount, the figure rises to approximately 41 GtCO₂e/year — roughly eight times the combined steel and cement total. Even the most conservative reading in this report places full-boundary forestry impact above the combined disclosed emissions of both sectors combined; the unadjusted Maxwell et al. figure, which this report does not adopt as its own estimate, illustrates the upper end of what the published literature’s own (regional) multiplier would imply if applied globally without the discount this report uses.

This is the asymmetry stated as a single comparison: the global steel and cement industries are each subject to IEA tracking, national emissions-trading-scheme coverage, CBAM border-carbon-adjustment exposure, and mandatory corporate disclosure regimes, for a combined direct emissions figure of roughly 4–5 GtCO₂e/year. Global forest loss and degradation, full-boundary accounted by a method already published and peer-reviewed five years before this report, produces a climate impact several times that size — and is not subject to an equivalent disclosure regime in any major jurisdiction’s GHG inventory or in the EPD/ISO 21930 standard that governs construction-material carbon claims.

Section 5 finding: The working hypothesis stated at the outset of this report — that full-boundary forestry carbon impact would exceed the emissions of major industrial sectors currently subject to far heavier scrutiny — is supported by the data, using a deliberately conservative reading of an already-published, peer-reviewed methodology. The comparison is not close. Even at the report’s own conservative floor, full-boundary forest-loss impact exceeds combined global steel and cement emissions by a factor of 3.5–6 $\times$.

SECTION 6

Addressing Two Specific Objections

Two objections are anticipated, specific, and answerable directly rather than left for a hostile reader to raise unaddressed.

6.1 The land-use counterfactual

A standard industry rebuttal to forgone-removals accounting is that, absent commercial timber demand, the land in question would not remain standing forest indefinitely — it would more likely be converted to agriculture, pasture, or development, each of which typically produces a worse carbon outcome than

continued forestry. This is a real and legitimate counterfactual question, and it deserves a direct answer rather than a footnote.

The answer is that this report's scope is deliberately narrower than that broader land-use debate. Maxwell et al. (2019), the method this report relies on, examined intact forest loss specifically — forest that was not already scheduled for conversion to another land use, and where the relevant counterfactual is continuation of the existing forest state, not conversion to an alternative use. This report inherits that same scope: its forgone-removals estimate answers the question "what does full accounting show for forest that was cleared or degraded, relative to that same forest remaining intact," not the separate question "what is the best available land-use policy for land currently in any given state." Where timber harvest is occurring on land that would, under a realistic regulatory and market counterfactual, otherwise be converted to non-forest use rather than remain standing, this report's forgone-removals figure would overstate the comparison for that specific hectare, and a land-use-conversion-adjusted estimate would be lower. This report does not attempt to estimate what share of the global harvest figure falls into each category, and Section 5's global comparison should be read with that scope limitation in mind. The Reagan Forestry Legacy companion paper's documentation of plantation-specific harvest economics (rather than land conserved from conversion) is the more precise lens for that narrower question.

6.2 Time-horizon sensitivity (GWP-20 vs GWP-100)

This report, consistent with the DRL Codex, uses IPCC AR6 GWP-100 values throughout. Because methane's radiative forcing is far more potent over a 20-year horizon (IPCC AR6 GWP-20 $\approx$ 81–83) than over 100 years (GWP-100 $\approx$ 27–30), and because the end-of-life methane pathway is one component of the direct-emissions figure used in this report's comparisons, the relative weight of that component — though not the larger forgone-removals term, which is a CO₂ flux unaffected by this choice — would be materially larger under a GWP-20 framing. The forgone-removals term, which is the dominant component of this report's headline figure (Section 4.3), is calculated in CO₂ mass directly from flux and sink data and is not time-horizon-sensitive in the way a methane-denominated figure would be. The report's overall comparison to steel and cement (Section 5) is therefore not substantially sensitive to the GWP-20/GWP-100 choice, because the term driving that comparison is a direct carbon-flux quantity rather than a methane-equivalence conversion. A dedicated sensitivity analysis applying GWP-20 throughout, including to the slash/decomposition methane pathway documented elsewhere in the DRL Codex, is identified here as warranted future work and is not performed in this report.

SECTION 6A

Further Limitations

In the interest of the same evidentiary standard applied throughout the DRL corpus, the following limitations are stated explicitly, in addition to those in Section 6.

This report does not claim a single, precise GWP-denominated figure for "cutting a forest." Section 2 explains why no such figure can exist under GWP's own definition. The carbon-flux component can be quantified (Sections 3–5); the evapotranspiration and albedo components cannot be converted to GWP-compatible units without contested modelling assumptions, and are therefore reported qualitatively, not folded into the headline figure.

The Maxwell et al. (2019) multiplier was derived for pantropical intact forest — high-biomass, high-sequestration-rate forest — and was not presented by its authors as a universal global coefficient. This report applies a conservative, self-imposed discount (2.0–3.0× rather than the published 6.26×), stated explicitly in Section 4.3 as a modelled estimate rather than a separately validated global figure, specifically to avoid overstating the case, since global forest loss includes substantial lower-biomass temperate and boreal secondary forest and plantation harvest with materially lower per-hectare sequestration rates than intact tropical forest.

This report does not allocate the global forestry figure to any single jurisdiction, company, or harvest practice. It establishes the scale of the global accounting omission; attribution to specific actors (the subject of the Reagan Forestry Legacy and Tree of Many Sales companion papers) is a separate analytical exercise with its own evidentiary requirements.

The albedo component (Section 2.3) is genuinely two-directional — cooling in snow-affected temperate/boreal zones, warming in tropical zones — and is not assumed to point in one direction in this report, consistent with the corpus-wide standard of disclosing complicating evidence rather than omitting it.

This report's forgone-removals estimate assumes a counterfactual of continued intact forest, consistent with the scope of the Maxwell et al. (2019) method it relies on. It does not estimate the share of global harvest occurring on land that would otherwise be converted to non-forest use under a realistic regulatory and market counterfactual; for that share, a land-use-adjusted estimate would be lower than this report's figure. See Section 6.1.

This report's headline comparison (Section 5) is calculated using IPCC AR6 GWP-100 throughout and has not been independently recalculated at GWP-20; Section 6.2 explains why the comparison is expected to be relatively insensitive to this choice, but a dedicated sensitivity analysis has not been performed.

Disclosed committee participation discussed in Section 0.2 is presented as documented configuration only. No claim of intent, coordination, or improper conduct is made or implied regarding any named individual, committee, or institution.

Section 0.6's account of the IPCC sector-boundary convention and its downstream reinterpretation is a historical-sequence claim, not a quantitative one, and does not affect the figures in Sections 3–5. It identifies that an explicit disclaimer exists in the public record and predates the standards examined in this report; it does not attempt to establish when, by whom, or through what specific intermediate steps the disclaimer was dropped between the 1996/2006 IPCC guidance and the 2011–2017 ISO/CEN standards, which would require a separate documentary review beyond this report's scope.

SECTION 7

The Structural Conclusion

Section 2 established that GWP, as a metric, was built to compare gas masses, and that a standing forest's climate function cannot be fully expressed in that unit. Sections 3 through 5 showed that even the portion of forest function that does convert cleanly to GWP-compatible units — carbon flux — is large enough, once the forgone-removals term is included using an already-published and peer-reviewed method, to place full-boundary forestry impact above the combined disclosed emissions of the global steel and cement industries.

The accounting convention responsible is not a forestry-specific error. It is the general rule that pre-extraction baseline equals zero, applied without exception across material categories. That rule is a reasonable approximation for ore, which does not perform an ongoing atmospheric removal function comparable to standing forest. It is not a reasonable approximation for standing forest, and the size of the resulting omission is not small: this report's own conservative, self-disclosed estimate places it at a minimum of roughly ten gigatonnes CO_{2e} per year beyond what conventional accounting recognises (Section 4.3), and Maxwell et al.'s own published pantropical rate, applied to the net sink without discount, places it well beyond that (Section 5) — though that unadjusted figure is presented for context, not adopted as this report's own estimate.

The same standard — "pre-extraction climate function is zero" — produces a true result when applied to a mine and a false result when applied to a forest. Current GWP-based material accounting does not adjust the rule for the case where it produces a false result. It applies the same zero to both, and the resulting omission, on the data assembled in this report, is larger than the entire disclosed annual carbon footprint of global steel and cement production combined.

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INTEGRATION NOTE

This report is offered as a standalone technical document and as a candidate source for a new Codex section on the structural limits of GWP-based material accounting. It does not modify any existing DRL finding; it adds a fourth, distinct line of analysis alongside the three existing liabilities (soil organic carbon efflux, end-of-life methane, foregone sequestration at the project/building level). Where this report's global-scale foregone-removals analysis and the Codex's building-level foregone-sequestration liability overlap conceptually, they remain analytically distinct: the Codex liability is computed per building per the Stephenson/Luyssaert NPV-discounted method; this report's figure is a global-scale, peer-reviewed-method comparison to industrial sectors, intended to establish the order of magnitude of the omission system-wide rather than per project.